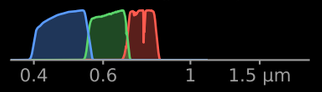
**DISCOVER · wide-field surveys find ~10³–10⁴ lensed AGN**

**CHARACTERIZE**

## Rubin LSST

pixel 0.2″ · PSF 0.71–0.77″

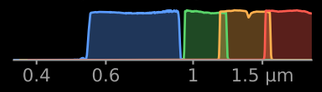
RGB = i r g · 10-yr




## Euclid

pixel 0.1–0.3″ · PSF 0.16–0.5″

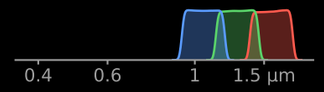
Q1 style: VIS + Y J H · wide




## Roman

pixel 0.11″ · PSF 0.087–0.13″

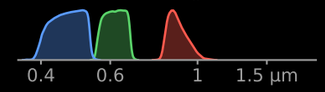
RGB = F158 F129 F106 · HLWAS



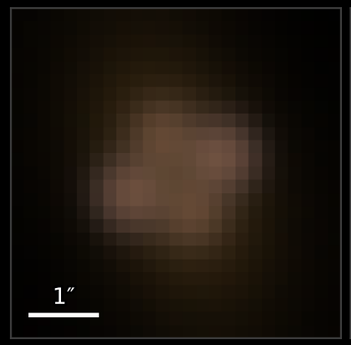

## Lazuli WCC

pixel 17 mas · PSF 33–62 mas

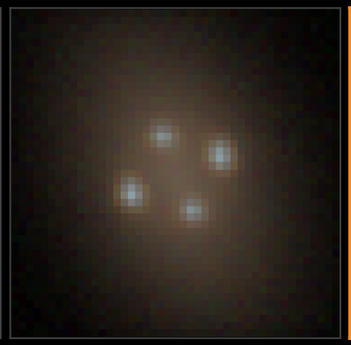
RGB = LZz LZr LZg · 4×300 s

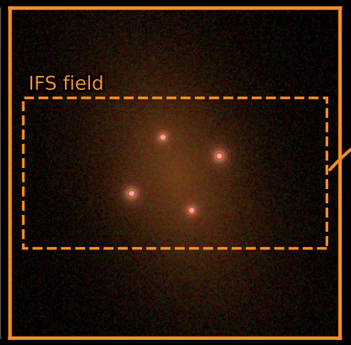



**Quad** $z_d$=0.43 $z_s$=1.87

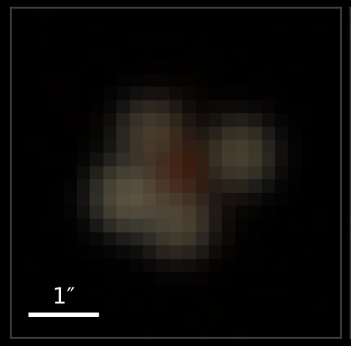




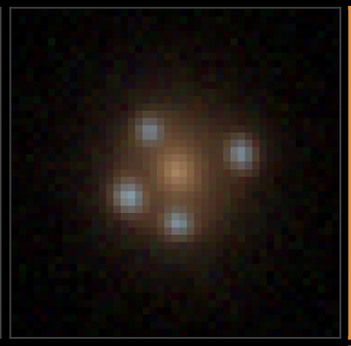



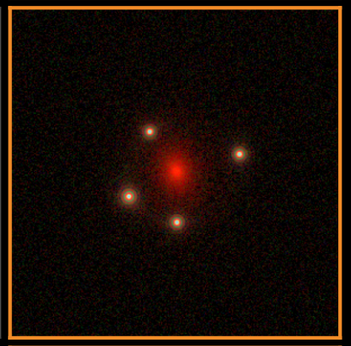



**Quad** $z_d$=0.89 $z_s$=2.74

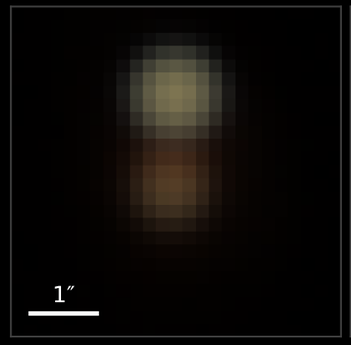


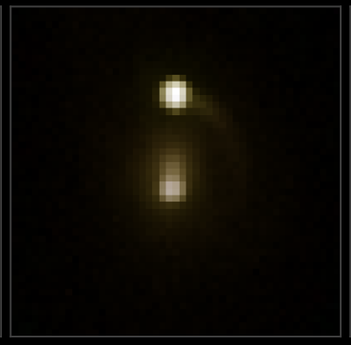


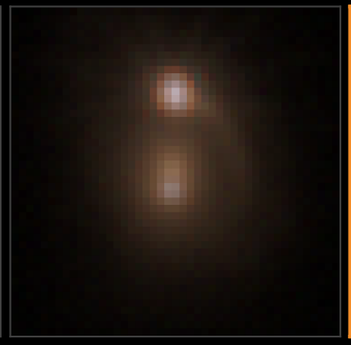





**Double** $z_d$=0.57 $z_s$=1.22



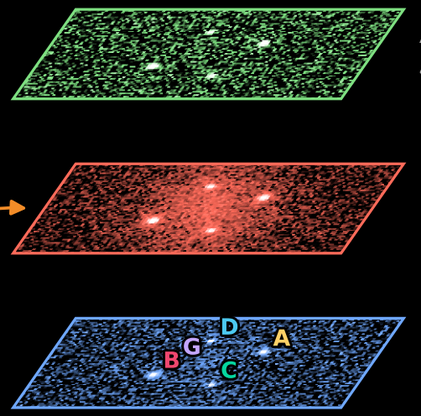







## LAZULI IFS · one pointing, every image

0.4–1.7 µm · R ≈ 100–500 · 40 mas spaxels · 4 × 900 s




**1.40 µm: Hβ line only**

rest 0.486 µm × (1 + $z_s$): quasar images only

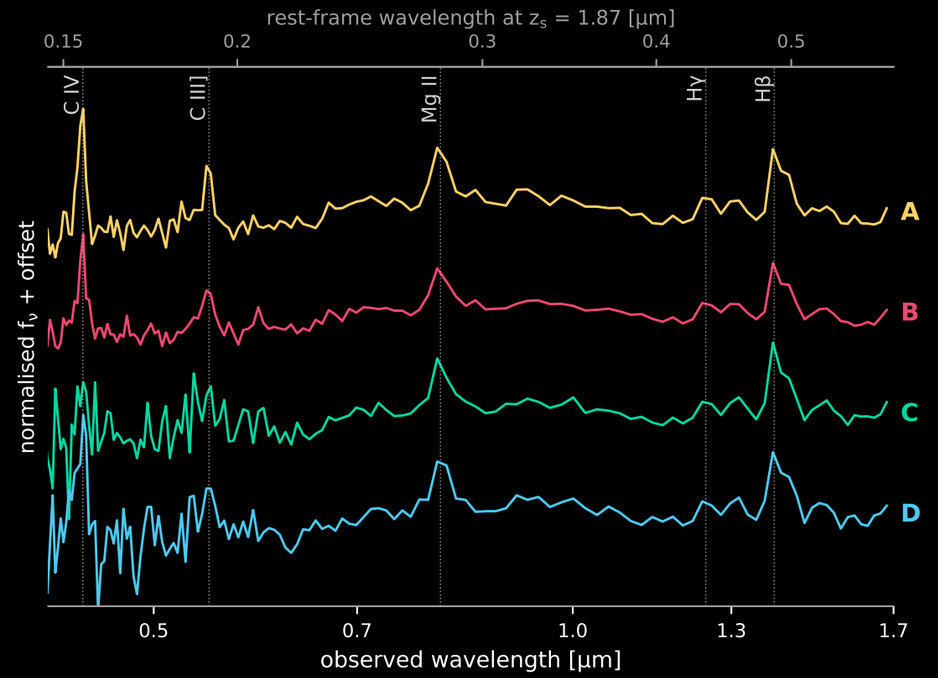
**1.00–1.20 µm continuum**

quasar images + lens galaxy

**0.58–0.76 µm continuum**

quasar images dominate

Same broad lines at the same redshift in every image → lensed quasar confirmed




Simulated with SLSim: COSMOS hosts, qsogen quasar SEDs, static microlensing. Lazuli: 3.06 m, Strehl > 0.8 at 633 nm; specs from Roy et al. 2026 (arXiv:2601.02556). Lazuli throughput, sky, noise and exposure times are notional. Lensed-AGN counts: Oguri & Marshall 2010.